

Adoption of AI-Based Workflow Automation Tools Among Vietnamese Teachers: Evidence From a Practice-First Workshop

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Abstract—Teachers are asked to do more every year. Much of that work is repetitive and invisible. Lesson preparation, forms, reports, and messaging quietly consume evenings and weekends. We report a four hour practice first professional development workshop for Vietnamese K–12 teachers that introduced AI assisted workflow automation using ChatGPT, Google Apps Script, and NotebookLM. We analyze an anonymous post workshop survey (203 responses; 194 complete for the main analysis). Teachers reported high perceived benefits and high intention to continue using the tools. Internal consistency is strong for Benefits, Support climate, and Intention. We test a UTAUT inspired post workshop intention model where Benefits and Support climate represent usefulness and facilitating conditions, while time availability and coding anxiety capture practical constraints. Benefits is the strongest predictor of Intention ($R^2 = 0.694$). Time availability adds incremental explanatory power ($\Delta R^2 = 0.017$). Capability outcomes also vary by baseline programming experience. We close with design guidance grounded in what teachers built, what they struggled with, and what support they asked for.

Index Terms—teacher professional development, workflow automation, generative AI, technology adoption, UTAUT, K–12 teachers, Vietnam, Google Apps Script, NotebookLM

I. INTRODUCTION

Many teachers already know what would help. Fewer repetitive clicks. Fewer copy paste steps. Fewer late night admin tasks. What blocks change is rarely motivation alone. It is time, confidence, and a support structure that survives after the workshop ends.

Generative AI can help teachers draft content quickly. Lightweight scripting can turn those drafts into repeatable workflows. Together, they can reduce friction for common tasks such as quiz creation, Google Form setup, spreadsheet reporting, and parent communication. Adoption research provides a useful lens for why teachers choose to continue using a tool after first exposure. In TAM and UTAUT style models, usefulness like beliefs and enabling conditions are reliable predictors of intention [1], [2].

This paper studies a workshop designed to meet teachers where they are. It was practical, template driven, and built around tasks that teachers recognized immediately. We report

what teachers said right after the session. We also test a theory informed post workshop intention model that matches what we measured in the survey.

A. Research Questions

RQ1. How do teachers rate Benefits, Support climate, and Intention right after the workshop?

RQ2. In a UTAUT inspired model, do Benefits and Support climate predict Intention?

RQ3. Do practical constraints (Time availability, Coding anxiety) add explanatory power beyond Benefits and Support climate?

RQ4. Are capability outcomes linked to baseline programming experience?

Contributions.

- A post workshop case study of a four hour hands on GenAI plus Apps Script training for Vietnamese K–12 teachers ($N = 203$).
- A tested UTAUT inspired post workshop intention model that separates perceived value from practical constraints.
- Training design guidance grounded in capability outcomes and participant feedback, including differentiated tracks and a lightweight follow up support structure.

II. RELATED WORK

TAM and UTAUT highlight perceived usefulness (or performance expectancy) and facilitating conditions as key determinants of intention [1], [2]. In teacher adoption settings, organizational support and self efficacy are often related to persistence, especially when new tools require experimentation [3]–[6].

Professional development research consistently emphasizes active learning, coherence with teachers’ needs, and opportunities to practice and collaborate [7]–[10]. This matters for AI workflows because the learning is not only conceptual. Teachers must troubleshoot, adapt templates, and decide when verification is required.

Our study is narrower on purpose. It focuses on workflow automation that combines AI assistance with simple scripting.



Fig. 1. Workshop setting (in person auditorium training).

It is also an intervention setting, not a general population survey. The goal is to understand immediate adoption signals after a practice first workshop, and to translate those signals into concrete training design choices [11]–[13].

III. METHODOLOGY

A. Workshop Context

The workshop was hosted in Vinh Long Province at the invitation of a university partner. The delivery team consisted of independent third party volunteers. The session lasted four hours in an auditorium setting. One presenter delivered the core instruction and live demos. Trainers circulated to help teachers during hands on tasks. The workshop goal was simple: reduce administrative workload by teaching a small set of reusable automations.

B. Workshop Modules and Templates

We built the workshop around three reusable automations plus one study support tool. Each piece was designed to be copied, tested, and adapted quickly.

- **Google Form generator script.** Teachers used ChatGPT to help draft Apps Script code, then generated Google Forms from a template script.
- **Grading sheet template.** Teachers used a spreadsheet template for grading aggregation and basic reporting.
- **Parent message template.** Teachers used a prompt and template structure for consistent parent communication.

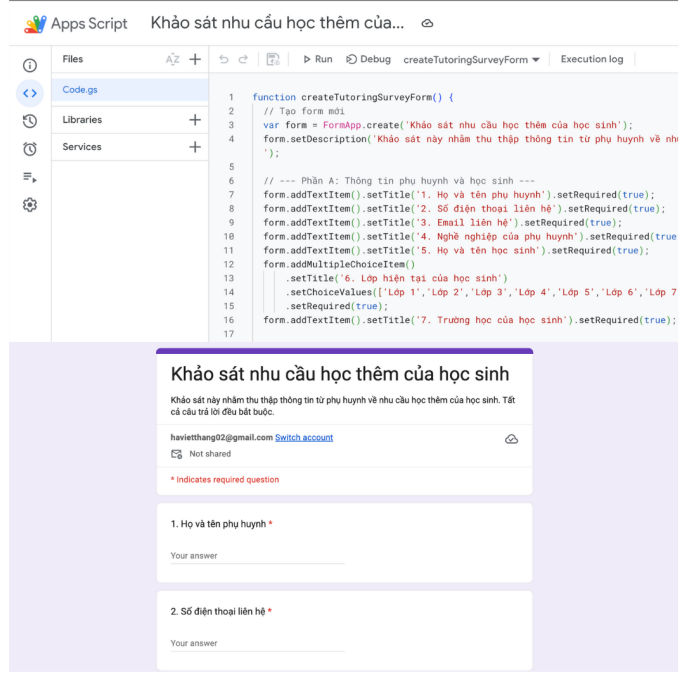


Fig. 2. Example workflow automation using Google Apps Script with AI assisted code generation for creating Google Forms.



Fig. 3. NotebookLM Video Overview used to transform lesson material into an audio summary.

- **NotebookLM study materials.** Teachers used NotebookLM to transform lesson documents into study artifacts, including Audio Overview.

We emphasized verification and safe use. We asked teachers to treat AI outputs as drafts, especially in high stakes contexts.

C. Participant Questions and Friction Points

During hands on time, most help requests were technical questions. The most common friction point was copy paste

failure during code transfer. Teachers reported that ChatGPT outputs can look similar across prompts. They also reported occasional bugs in generated code. These issues are small on paper, but they can stall novices quickly. Trainers therefore focused on debugging habits, small test steps, and known failure patterns.

D. Follow Up Support

After the workshop, we supported participants through a Zalo group chat. We used it to share updates on new AI tools, templates, and quick fixes. The intent was to keep momentum alive with a low friction channel.

E. Participants and Survey

Participants were in service teachers. Teaching levels were primary (27.6%), middle (31.5%), and high school (40.9%). Median teaching experience was 17 years.

The survey was anonymous and voluntary. It was presented in Vietnamese. It used a four point Likert scale with no neutral option (1 = strongly disagree, 4 = strongly agree) [14], [15]. Participants saw a short consent statement before responding. Attendance was assigned through schools, and teachers received certificates and compensation. That setting can increase positive ratings, so we interpret results as immediate post workshop signals rather than long term outcomes.

F. Measures

We built the measures around what teachers needed to decide after the workshop. Is this helpful. Is support present. Will I keep using it.

1) *Multi item scales*: We computed mean scores for three scales. Higher scores mean stronger endorsement.

- **Benefits (usefulness like)**. Five items (Q14, Q24 to Q27). $\alpha = 0.913$.
- **Support climate (facilitating conditions like)**. Four items (Q15, Q18 to Q20). $\alpha = 0.864$.
- **Intention**. Two items (Q28 to Q29). $\alpha = 0.893$.

We report Cronbach's α as internal consistency [16]. We do not treat it as proof that a scale is one dimensional [17], [18].

2) *Practical constraint indicators*: We treat two practical constraints as separate indicators.

- **Time availability**. Q21.
- **Coding anxiety**. Q17 [19].

We also report Confidence for scripting (Q16) descriptively to characterize the room.

3) *Capability outcome*: We used "scripts created after today" (Q11) as a simple capability outcome. We tested its association with baseline programming experience (Q7).

G. Analysis Plan

We report descriptive statistics and internal consistency [16]. We then test a UTAUT inspired post workshop intention model [20]. Benefits reflects usefulness like beliefs and Support climate reflects facilitating conditions. Practical constraints are modeled as time availability and coding anxiety [1], [2]. We estimate hierarchical regression models. Model 1 includes

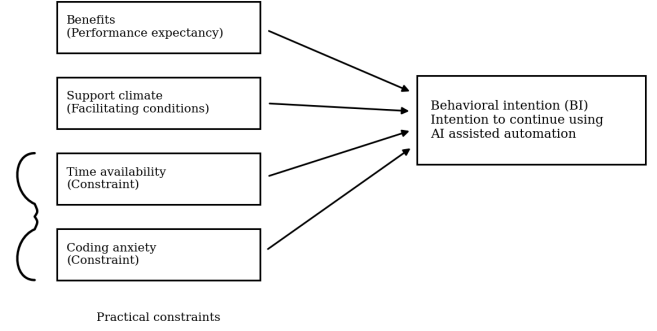


Fig. 4. Conceptual model for post workshop intention to adopt AI assisted workflow automation.

TABLE I
DESCRIPTIVE STATISTICS FOR KEY SCALES AND INDICATORS ($n = 194$)

Construct	Type	Mean (SD)	95% CI
Benefits (usefulness like)	5 item scale	3.17 (0.39)	[3.11, 3.22]
Support climate	4 item scale	3.13 (0.41)	[3.07, 3.18]
Intention	2 item scale	3.11 (0.41)	[3.06, 3.17]
Time availability	single item	3.04 (0.51)	[2.96, 3.11]
Coding anxiety	single item	2.93 (0.60)	[2.85, 3.02]
Confidence for scripting	single item	3.09 (0.49)	[3.02, 3.16]

Note: Likert responses were coded 1 to 4 from Vietnamese labels.

Benefits and Support climate. Model 2 adds Time availability and Coding anxiety. We check multicollinearity using VIF [21]. For capability, we use chi square tests and Cramer's V.

IV. RESULTS

A. Descriptive Statistics and Reliability

Table I summarizes scales and key indicators ($n = 194$). Means cluster near the upper bound. This suggests a ceiling effect, which is plausible in a post workshop setting with compensation and certificates. Table II reports internal consistency for multi item scales.

B. What Teachers Reported They Could Automate

Teachers selected multiple tasks they felt able to automate after the session. The most common were creating and sending Google Forms (121), creating teaching materials or lesson plans (115), and sending notification emails (93). Fewer selected automatic grading (56), attendance or timetable support (49), and score aggregation or analysis (48). This pattern matches the workshop structure. Teachers first learned workflows that create visible outputs quickly. More complex data workflows were less common in a single session.

C. Regression: Predictors of Intention

We estimated two hierarchical regression models with list-wise deletion ($n = 194$). Model 1 includes Benefits and Support climate. Model 2 adds Time availability and Coding anxiety.

TABLE II
INTERNAL CONSISTENCY (CRONBACH’S α) FOR MULTI ITEM SCALES

Scale	Cronbach’s α
Benefits	0.913
Support climate	0.864
Intention	0.893

Note: Indicators are single items. α is not applicable.

TABLE III
HIERARCHICAL REGRESSION PREDICTING INTENTION ($n = 194$)

Predictor	Model 1 β	Model 2 β
Benefits	0.679***	0.666***
Support climate	0.179*	0.077
Time availability	–	0.194**
Coding anxiety	–	-0.054
R^2	0.694	0.711
ΔR^2	–	0.017
Model F	216.48	116.34

Note: Standardized coefficients. * $p < .05$, ** $p < .01$, *** $p < .001$. In Model 2: $p_{\text{Support}} = 0.393$, $p_{\text{Time}} = 0.001$, $p_{\text{Anxiety}} = 0.322$. Max VIF = 5.25.

Benefits is the strongest predictor of Intention. Time availability adds a smaller but meaningful increment. Support climate and coding anxiety do not contribute unique explanatory power once Benefits and Time availability are included. In this dataset, the practical bottleneck is not whether teachers can see value. It is whether they can find time to apply the workflows after the session.

D. Capability Association

We tested whether “scripts created after today” (Q11) is associated with baseline programming experience (Q7). The association was significant: $\chi^2 = 34.50$, $df = 12$, $p = 0.00056$, with Cramer’s $V = 0.242$ ($N = 196$). This is a small to moderate association. It points to a simple design lesson. Teachers start from different baselines, and a single track workshop will not fit everyone equally well.

V. DISCUSSION

The story here is human, not technical. Teachers want relief from repetitive work. When they can see an immediate win, they lean in. In our workshop, wins were concrete. A form appears. A grading sheet works. A message template reduces effort. An Audio Overview turns a lesson document into a study artifact.

Our results show high Benefits and high Intention right after a single session. This aligns with technology acceptance research that emphasizes usefulness like beliefs as a driver of intention [1], [2]. The strongest predictor of Intention is Benefits. This matters because it suggests teachers did not merely enjoy the session. They believed the workflows would save time or reduce mistakes.

At the same time, time remains the constraint that separates motivation from follow through. Time availability adds

explanatory power beyond Benefits and Support climate. In practice, this often looks like a teacher who wants to adopt, but postpones the first reuse because the next week is full of grading, meetings, and reports.

Support climate does not add unique variance once Benefits and Time availability are included. We do not interpret this as “support does not matter.” Instead, it suggests a narrower conclusion for this context. Immediately after training, teachers may interpret supportive climate through the same lens as usefulness and available time. In other words, support matters most when it protects time, reduces troubleshooting cost, and provides ready templates.

A. Implications for Training Design

Our workshop design already leaned toward active learning. Participants built artifacts during the session. This aligns with core features of effective teacher professional development, especially active learning and coherence with classroom needs [7]–[9]. The results point to three practical upgrades.

1) Differentiated tracks. Capability outcomes vary by baseline programming experience. A two track structure is a low cost fix.

Track A: Template first. Focus on reuse. Teach safe edits. Teach how to change parameters without breaking a script. Provide a short checklist for common errors.

Track B: Builder. Focus on debugging and maintenance. Teach how to adapt a template across subjects and policies. Use small tasks that force logic changes.

2) Debugging scaffolds for copy paste failures. Teachers reported that copy paste bugs and near duplicate AI outputs slow them down. This is predictable in AI assisted code generation. The training response is also predictable. Encourage small test steps. Provide known good snippets. Teach how to locate the smallest failing line. Provide a simple error glossary.

3) Support that reduces time cost. Time availability predicts intention beyond perceived value. Follow up support should therefore be designed to save time, not only to motivate. A Zalo group chat can work well when it delivers template updates, quick fixes, and short office hour windows. A shared template library also helps teachers restart fast.

B. Threats to Validity

This is a post only design. It limits causal claims. It can inflate agreement due to social desirability and common method bias [22]. The forced choice scale can compress variance. That can weaken estimates for some predictors. Generalizability is limited because attendance was assigned and the workshop was delivered in one context.

Future work should add a delayed follow up and a behavioral indicator of reuse. Even a lightweight measure, such as counts of template reuse or form generation, would strengthen claims about adoption beyond intention.

VI. CONCLUSION

This study reports a practice first workshop that introduced Vietnamese K–12 teachers to AI assisted workflow automation using ChatGPT, Google Apps Script, and NotebookLM.

Teachers reported high perceived Benefits and high Intention immediately after the session, and the tested UTAUT inspired model shows that Benefits is the strongest predictor of Intention, with Time availability adding incremental explanatory power beyond perceived value. In practical terms, teachers can be convinced and still struggle to follow through if they cannot find time to apply workflows during a normal week. Capability outcomes also vary by baseline programming experience, which supports a two track training design: a template first track focused on safe reuse and troubleshooting, and a builder track focused on adapting and maintaining scripts. A lightweight follow up channel such as a Zalo group should be treated as part of the intervention because it reduces the time cost of reuse and helps early wins become routine practice.

APPENDIX A SURVEY ITEMS USED IN THIS PAPER

Items were presented in Vietnamese. The text below is a short English gloss for readability.

A. Benefits (5 items)

- Q14. Automation reduces workload.
- Q24. Many tasks become easier or faster with automation.
- Q25. Automation gives more time for students.
- Q26. Automation improves accuracy and reduces errors.
- Q27. Automation benefits students.

B. Support climate (4 items)

- Q15. Teachers should learn automation.
- Q18. School leadership expects teachers to develop technology solutions.
- Q19. The school provides support and resources.
- Q20. I have colleagues or a community that supports learning technology.

C. Intention (2 items)

- Q28. I plan to develop or use more workflows.
- Q29. I will encourage other teachers to explore automation.

D. Constraint indicators

- Q21. I have enough time to experiment and develop automation.
- Q17. Writing code or scripts makes me hesitant.
- Q16. I feel confident learning and using scripts or code.

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